

Employee Churn Prediction Model Report

Data Preprocessing and Feature Engineering

The dataset was preprocessed by encoding categorical variables, scaling numerical features, and addressing class imbalance using SMOTE. Feature engineering was conducted to select relevant features. The highly correlated features were identified and considered during model building to avoid multicollinearity issues.

Model Selection and Tuning

A Random Forest model was selected for its robustness and ability to handle high-dimensional data. Hyperparameter tuning was performed using grid search to optimize the model. The best parameters found were: `n_estimators=300`, `max_depth=None`, `min_samples_split=2`, `min_samples_leaf=1`, `bootstrap=False`.

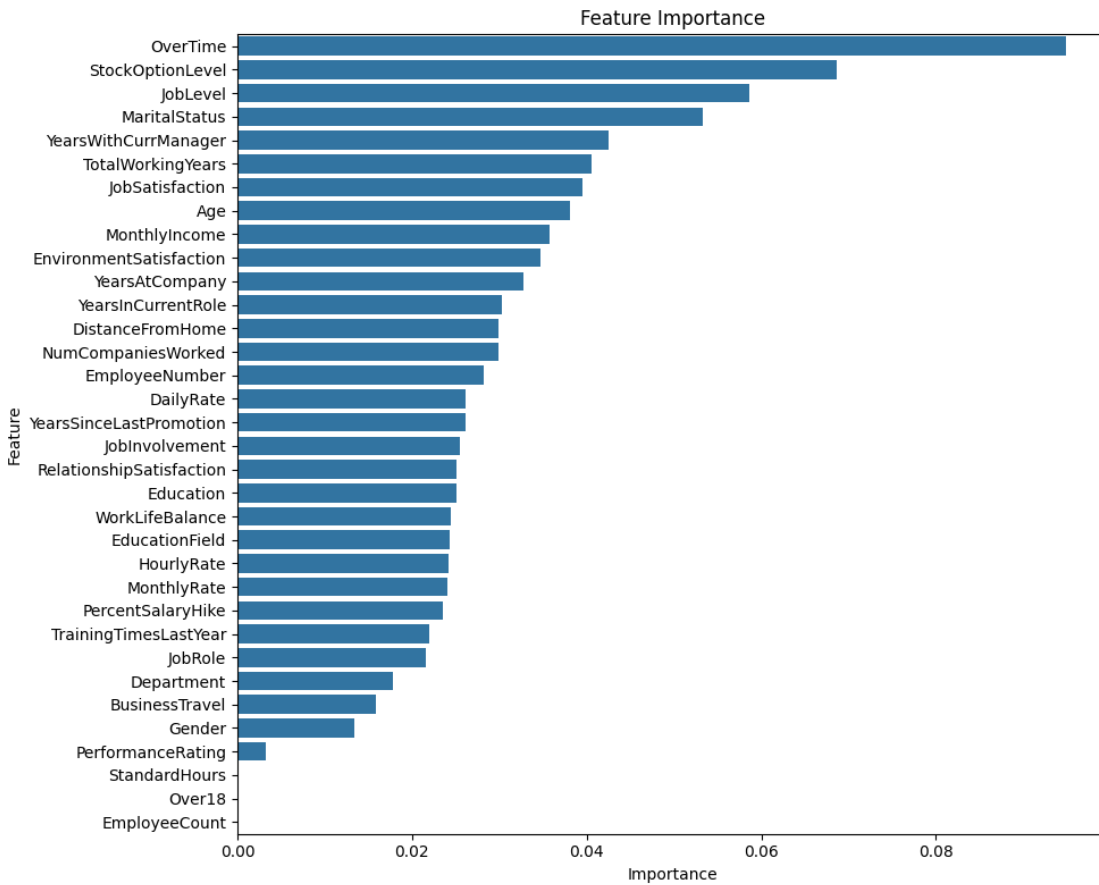
Evaluation Results

The model was evaluated using accuracy, precision, recall, F1-score, and ROC-AUC. The results indicate a high level of performance, with an accuracy of 93.32% and a ROC-AUC of 97.81%. The confusion matrix and classification report provide further insights into the model's performance.

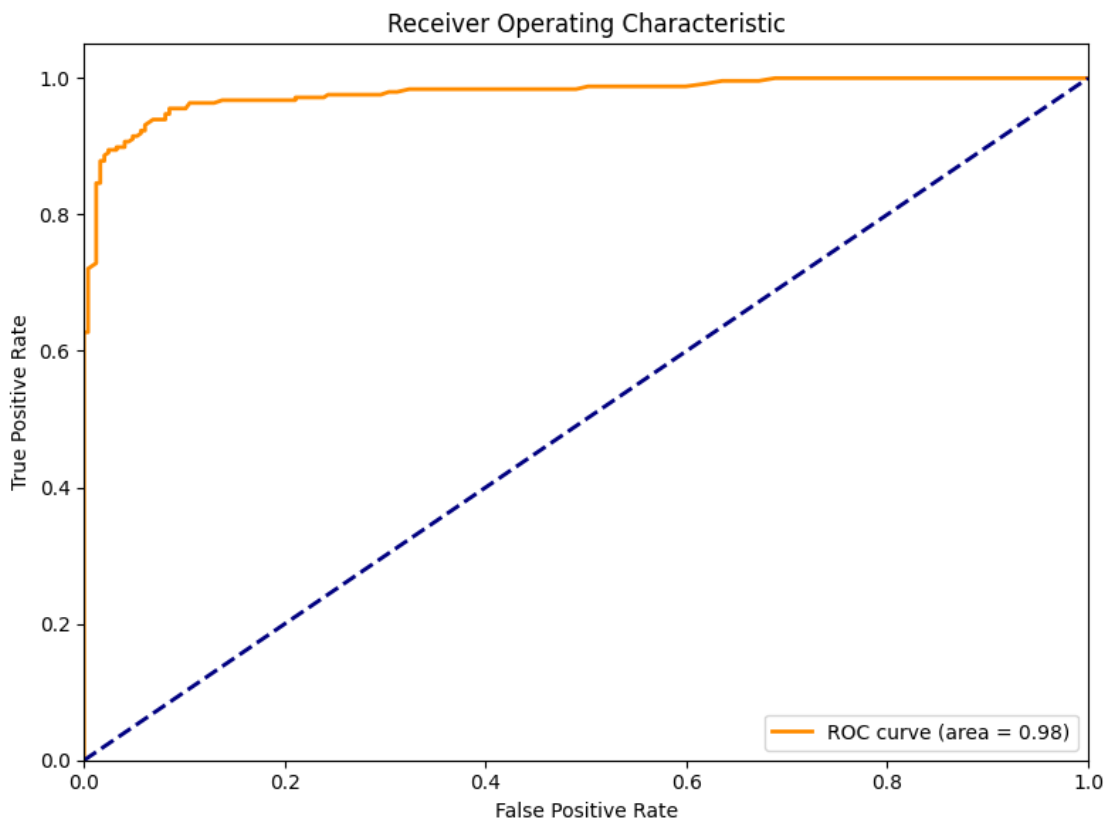
Visualizations

Feature Importance:

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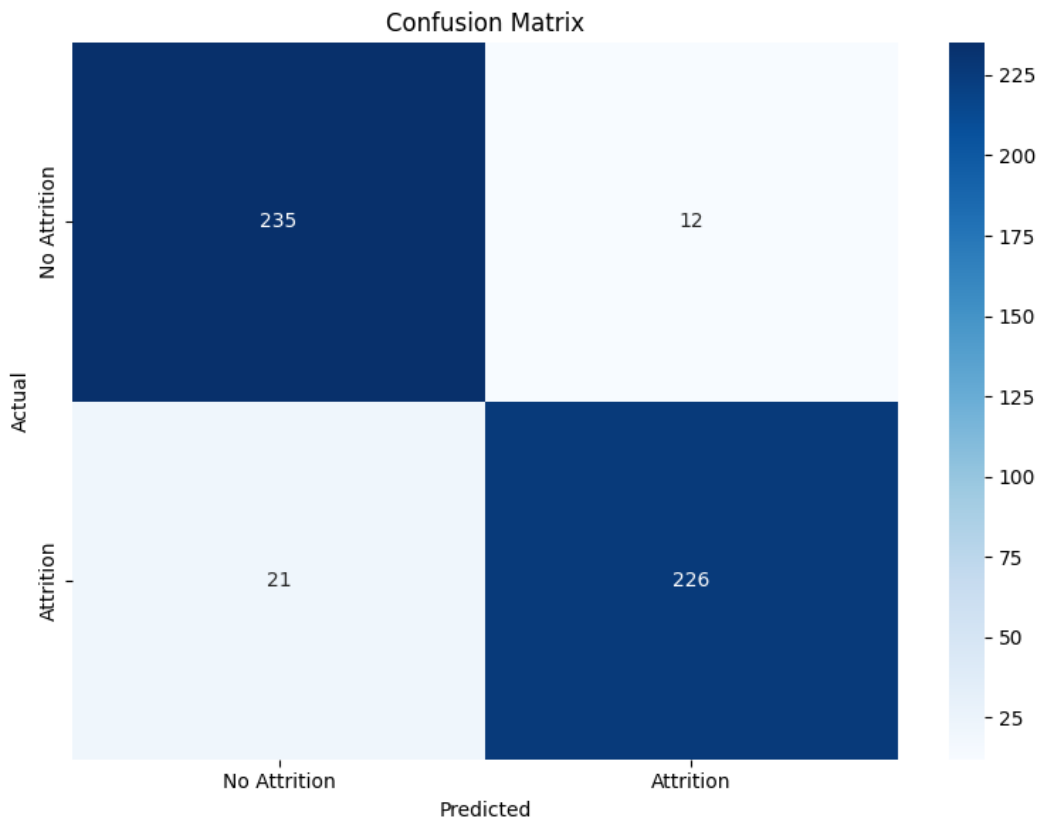


ROC Curve:



Confusion Matrix:

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Business Implications

The model can help the HR department identify employees at risk of attrition, allowing for targeted interventions to improve retention. This can lead to reduced turnover costs and a more stable workforce. By understanding the key factors contributing to attrition, such as job satisfaction and work-life balance, the company can implement strategies to enhance employee engagement and satisfaction.